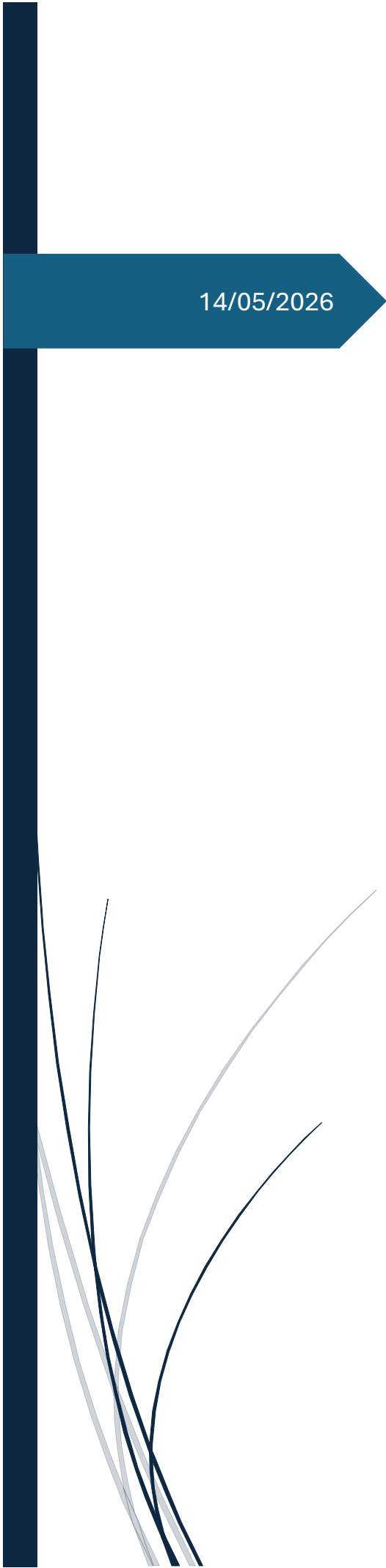




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PHQ/GAD score distribution analysis

Dataset :- PHQ-9/GAD-7

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DATA ANALYTICS INTERN

Introduction

Mental health issues such as anxiety and depression have become increasingly common among young adults, particularly among students and social media users. This project focuses on analysing the distribution of mental health assessment scores using demographic variables such as gender and age.

The analysis was conducted using an open mental health dataset containing demographic, behavioural, and psychological assessment information. The primary objective of this analysis is to understand how anxiety and depression scores vary across different genders and age groups.

Two standard mental health assessment scales were used:

1. **GAD-7 (Generalized Anxiety Disorder-7)** for anxiety assessment
2. **PHQ-9 (Patient Health Questionnaire-9)** for depression assessment

The report includes:

- a) Dataset overview
- b) Data cleaning and preprocessing
- c) Descriptive statistics
- d) Distribution analysis
- e) Box plots and violin plots
- f) Gender-wise analysis
- g) Age-wise analysis
- h) Key insights and findings

Objectives of the Analysis

The main objectives of this project are:

1. Analyse anxiety score distributions using GAD-7 scores.
2. Analyse depression score distributions using PHQ-9 scores.
3. Compare score distributions across gender.
4. Compare score distributions across age groups.
5. Identify variability, spread, and outliers in mental health scores.
6. Use professional visualisations such as box plots and violin plots.

7. Generate insights that can support mental health awareness and research.

Dataset Overview

The dataset used in this analysis contains social media and mental health related information.

Dataset Summary

Attribute	Attribute
Total Records	Total Records
Gender Categories	Gender Categories
Age Range	Age Range
Anxiety Metric	Anxiety Metric
Depression Metric	Depression Metric

Important Columns Used

Column Name	Description
Age	Age of the participant
Gender	Gender category
GAD 7 Score	Anxiety assessment score
PHQ 9 Score	Depression assessment score

Data Preprocessing

Before performing analysis, the dataset was cleaned and prepared.

Preprocessing Steps

- a) Loaded dataset using Pandas.
- b) Checked dataset dimensions and structure.
- c) Identified missing values.
- d) Removed duplicate records.
- e) Selected relevant columns.
- f) Converted columns into proper data types.
- g) Removed invalid or null values.
- h) Created age groups for comparative analysis.

Age Grouping

Age	Group
18–19	Young Group
20–21	Adult Group
22+	Senior Student Group

Descriptive Statistics

GAD-7 Anxiety Scores

Statistic	Value
Mean	7.35
Median	7
Minimum	0
Maximum	21
Standard Deviation	4.63

PHQ-9 Depression Scores

Statistic	Value
Mean	5.38
Median	5
Minimum	0
Maximum	23
Standard Deviation	4.82

Gender Distribution

Gender	Count
Female	4028
Male	3972

The dataset is relatively balanced in terms of gender representation.

Distribution Analysis

Distribution analysis helps understand:

- a) Central tendency
- b) Variability
- c) Spread of scores
- d) Presence of outliers
- e) Density concentration
- f) Skewness of mental health scores

The following visualisations were used:

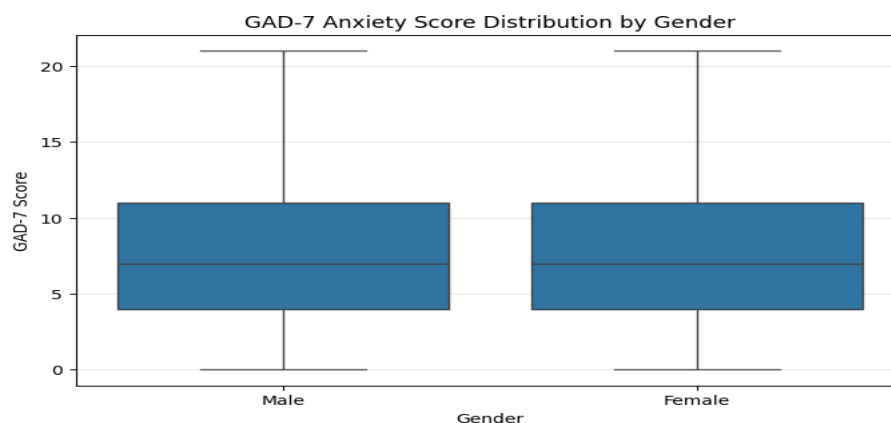
- a) Histogram plots
- b) KDE density plots
- c) Box plots
- d) Violin plots
- e) ECDF plots

Gender-wise Anxiety Score Analysis (GAD-7)

Box Plot Interpretation

The box plot comparing GAD-7 scores across gender showed:

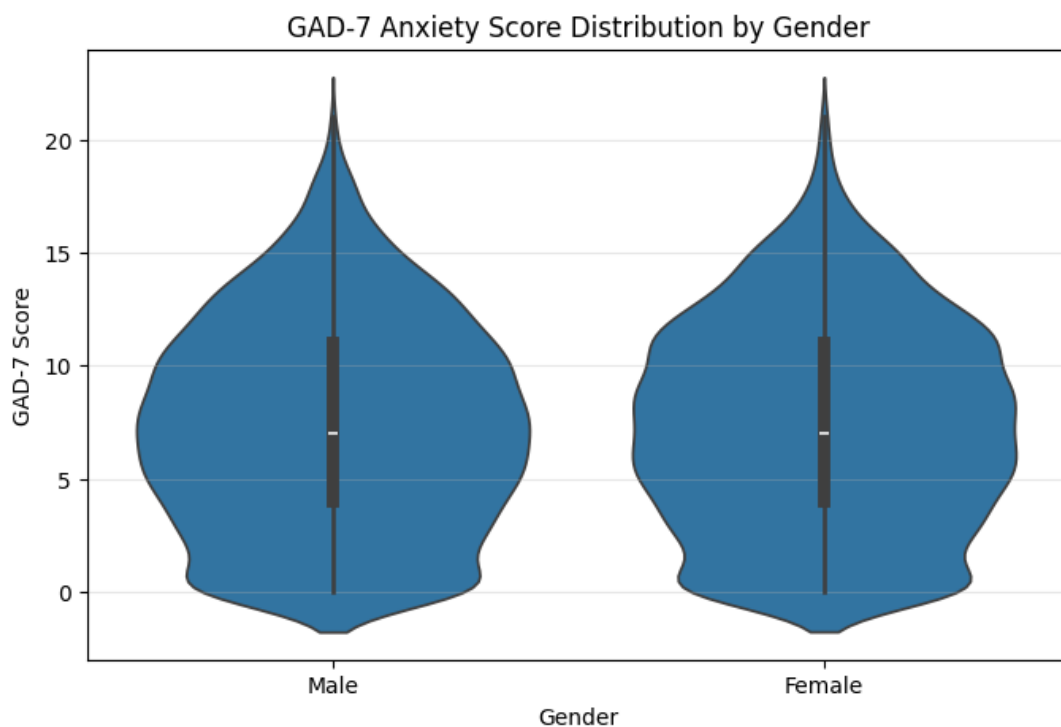
- a) Moderate variation in anxiety scores.
- b) Presence of outliers with high anxiety levels.
- c) Similar median anxiety scores between genders.
- d) Slightly wider score distribution among female participants.



Violin Plot Interpretation

The violin plot revealed:

- a) Dense concentration of anxiety scores in the lower-middle range.
- b) Broader spread in female participant distributions.
- c) Higher variability in some participant groups.



Key Findings

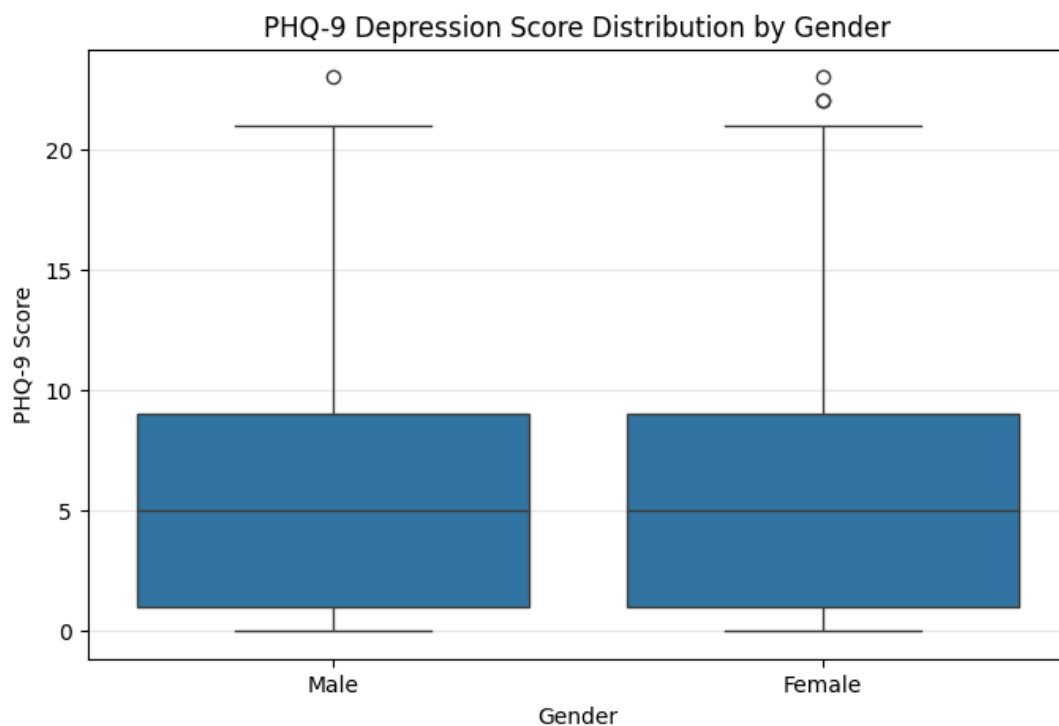
- I. Anxiety levels are moderately distributed across the population.
- II. Some individuals show significantly high anxiety scores.
- III. Distribution patterns indicate mental health variability among users.

Gender-wise Depression Score Analysis (PHQ-9)

Box Plot Interpretation

The PHQ-9 box plot showed:

- a) Wider distribution compared to anxiety scores.
- b) Presence of multiple high-score outliers.
- c) Moderate median depression levels.
- d) Comparable depression distributions across genders.



Box Plots

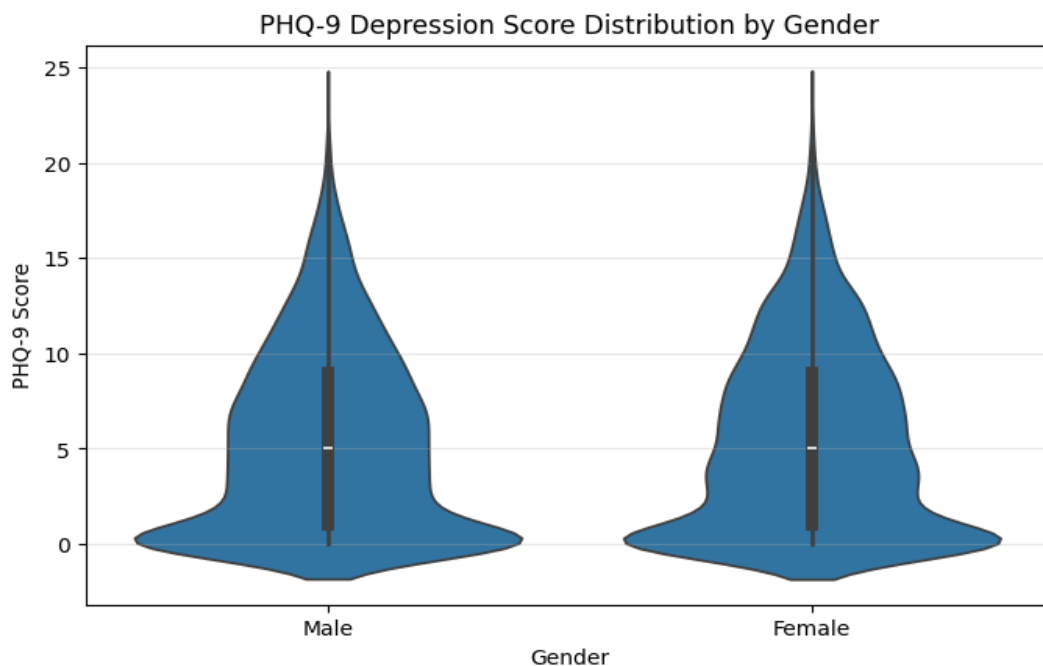
Box plots helped identify:

- a) Median values
- b) Interquartile ranges
- c) Outliers
- d) Variability between groups

Violin Plot Interpretation

The violin plot demonstrated:

- a) Greater spread in depression score density.
- b) Clustering of users around lower and moderate depression ranges.
- c) Presence of participants with severe depression scores.



Key Findings

- I. Depression levels vary significantly across users.
- II. Some participants display high PHQ-9 scores.
- III. Distribution indicates mental health risk variation among individuals.

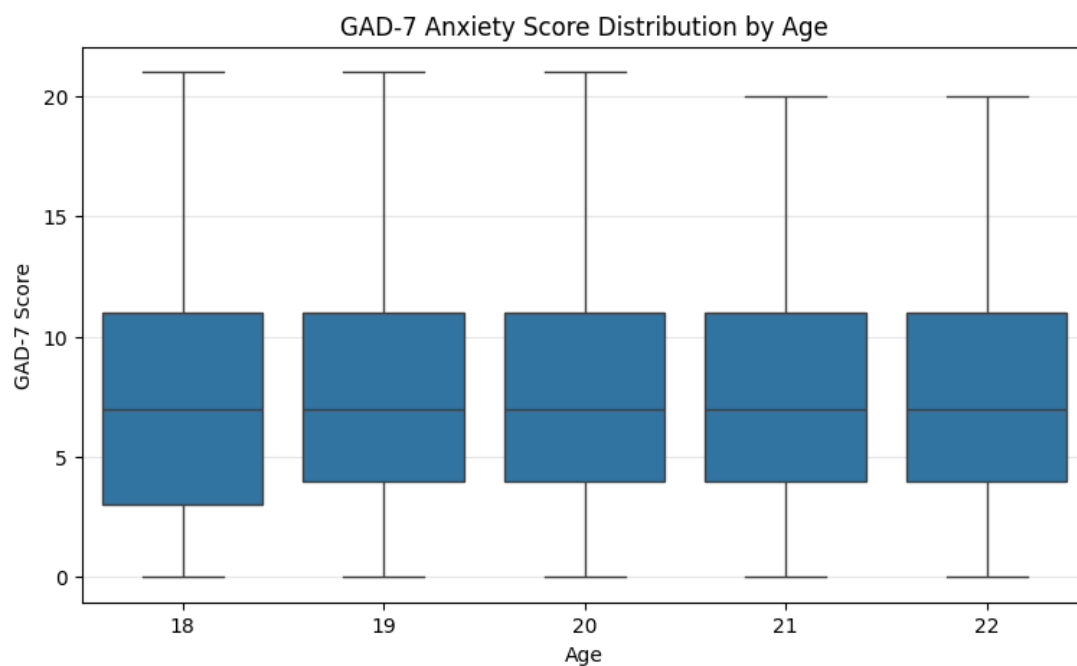
Age-wise Anxiety Score Analysis

The age-based GAD-7 distribution analysis showed:

- a) Anxiety scores vary across age categories.
- b) Younger participants demonstrated wider score variability.
- c) Median anxiety levels remained relatively stable across ages.
- d) Outliers were observed in all age groups.

Observations

- a) Ages 18–19 showed larger variability in some plots.
- b) Higher anxiety extremes were present among younger users.
- c) Score density was concentrated around moderate anxiety ranges.



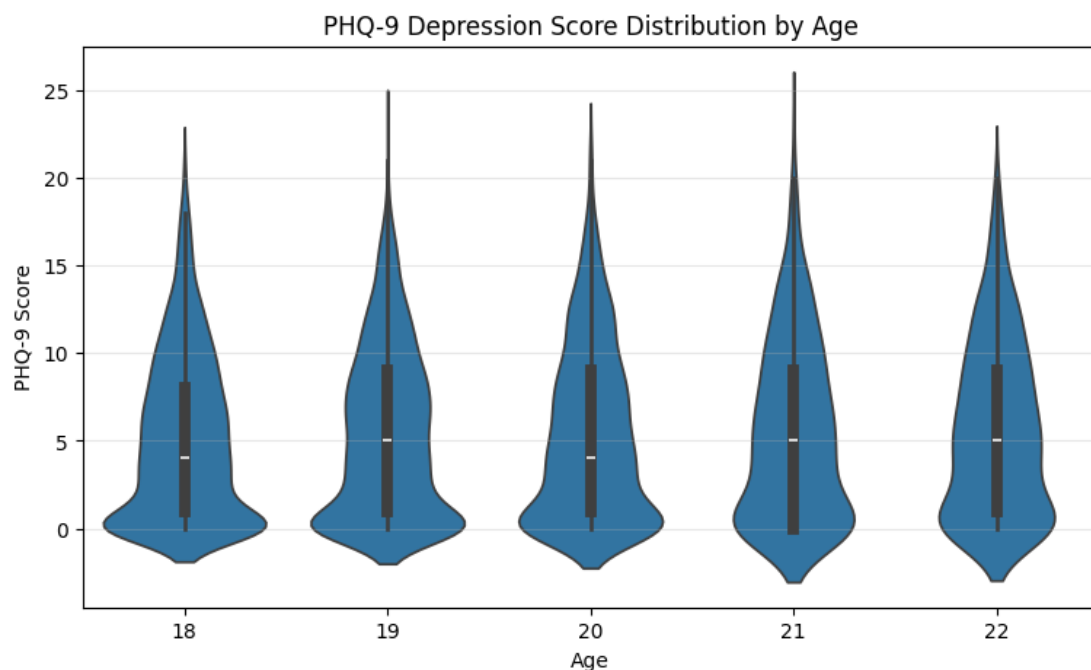
Age-wise Depression Score Analysis

The PHQ-9 age-based analysis indicated:

- a) Depression score distributions differ slightly across age groups.
- b) Moderate depression levels were common.
- c) Some age groups showed broader distribution tails.
- d) High-score outliers existed across all age categories.

Observations

- a) Depression variability increased in some younger age segments.
- b) Density plots showed clustering around low-to-medium depression scores.
- c) Extreme values suggest mental health concerns for certain participants.



Violin Plots

Violin plots provided:

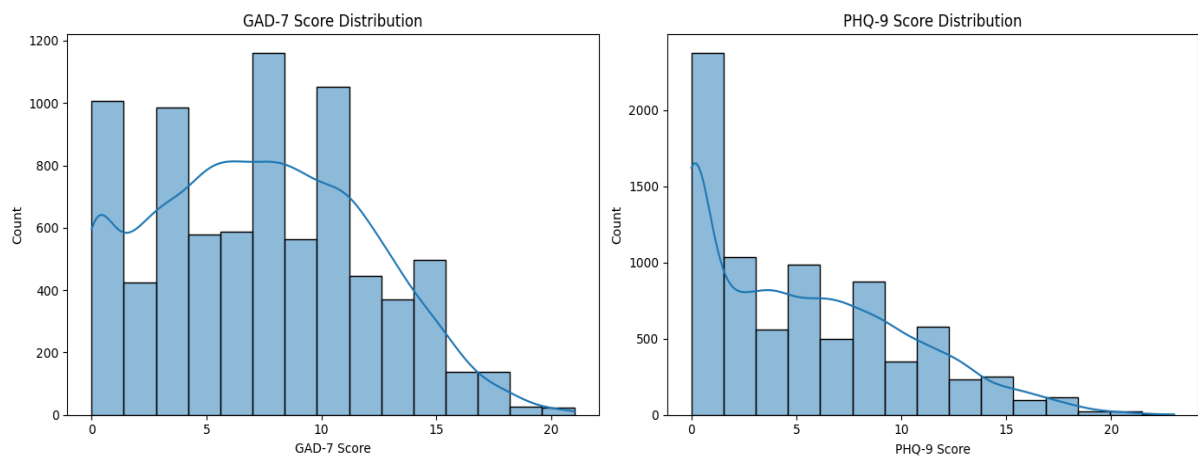
- a) Distribution shape
- b) Density structure
- c) Spread comparison
- d) Probability concentration

Distribution Visualisations Used

Histograms

Histograms were used to:

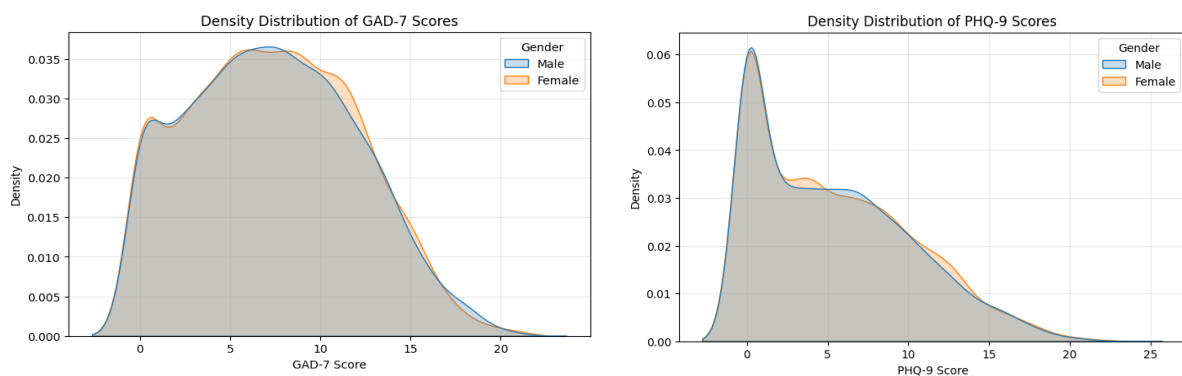
- Display frequency distribution
- Identify skewness
- Observe score concentration
- Detect high-frequency score ranges



KDE Density Plots

KDE plots provided:

- Smooth distribution curves
- Better understanding of density concentration
- Comparison across demographic categories



Key Insights

Major Findings

1. Anxiety and depression scores are moderately distributed within the dataset.
2. Several participants exhibit high anxiety and depression scores.
3. Female participants showed slightly wider score variability.
4. Younger participants displayed greater variability in some mental health measures.
5. Outliers indicate individuals potentially experiencing severe mental health symptoms.
6. Most users fall within low-to-moderate score ranges.
7. Distribution plots suggest mental health patterns vary across demographics.

Importance of the Analysis

This analysis can support:

- a) Mental health awareness initiatives
- b) Student wellbeing studies
- c) Psychological behaviour research
- d) Social media impact assessment
- e) Public health analytics
- f) Predictive mental health modelling

The visualisations provide valuable insights for:

- a) Researchers
- b) Healthcare professionals
- c) Mental health practitioners
- d) Educational institutions
- e) Data analysts

Limitations

Although the analysis provides useful insights, several limitations exist:

1. The dataset covers a limited age range.
2. Gender categories are limited.
3. Results are based on self-reported scores.
4. External environmental factors were not analysed.
5. Clinical diagnosis cannot be inferred directly from the dataset.

Future Improvements

Future analysis can include:

- a) Correlation analysis with screen time and sleep duration
- b) Predictive modelling using machine learning
- c) Clustering of mental health patterns
- d) Time-series behavioural analysis
- e) Sentiment analysis integration
- f) Risk prediction systems
- g) Interactive Power BI dashboards

Conclusion

This project successfully analysed the distribution of anxiety and depression scores across gender and age groups using professional statistical visualisations.

The findings demonstrate that mental health scores vary across demographic categories and that several users exhibit elevated anxiety and depression levels. Box plots and violin plots effectively highlighted variability, density concentration, and outlier behaviour within the dataset.

The analysis provides meaningful insights into mental health distribution patterns and can serve as a strong foundation for advanced mental health analytics, predictive modelling, and public health research.

Technologies Used

Tool	Purpose
Python	Data analysis
Pandas	Data preprocessing
NumPy	Numerical operations
Matplotlib	Visualisation
Seaborn	Statistical plotting
Jupyter Notebook	Analysis environment

Final Outcome

The project successfully achieved:

- a) Distribution analysis of GAD-7 and PHQ-9 scores
- b) Gender-wise comparison
- c) Age-wise comparison
- d) Professional box plot visualisation
- e) Professional violin plot visualisation
- f) Statistical insight generation
- g) Exploratory mental health analytics

This report demonstrates the practical application of data analytics in understanding mental health trends using demographic and psychological assessment data.